

Canonical Correlation Analysis of EEG for Classification of Motor Imagery

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Abstract—The performance of classification of various mental states using Electroencephalography (EEG) is often limited by the lack of information regarding the most discriminative channels and frequency bands. The paper proposes a Canonical Correlation Analysis (CCA) of EEG recorded during bilateral imagined hand movement. CCA determines linear transformation of EEG that is maximally correlated with a transformed version of neural response to imagined movement. In the proposed method, the linear weights are identified for the spatial and spectral components of EEG signal. The study investigates how the CCA-based transformation of the signal improves discrimination between two movement classes. Further, two parallel CCA blocks, determine two transformations of EEG, one that maximizes correlation with the corresponding class, and other that minimizes the correlation with the mismatched class. Features derived using this approach are used for single-trial classification of bilateral imagined hand movement. Time-frequency patterns of EEG derived from the proposed approach illustrate their discriminative ability in mu and beta bands. An average classification accuracy of 61.12% (109 subjects) and 73.34% (Best 40 subjects) are obtained. The results indicate the scope of CCA to obtain time-frequency representations of EEG and for single-trial classification of motor imagery (MI).

Keywords—Electroencephalography, Brain Computer Interface, Canonical Correlation Analysis

I. INTRODUCTION

Electroencephalography (EEG) is a non-invasive method of monitoring the electrophysiological activity of the brain. EEG owing to its simple and inexpensive setting, is widely used for neuroscience and neural signal processing research and clinical applications [1]. Brain Computer Interface (BCI) technology has been introduced with the goal of providing augmentative communication and control for those with severe neuromuscular disorders [1-3]. The BCI research based on EEG has investigated activation patterns and their modulations related to imagined bilateral motor movements, with the aim of developing a BCI based motor control system for the motor-disabled. Many researchers have successfully demonstrated the ability of scalp-recorded EEG as the basis of BCI, to obtain control signals without involvement of muscular control [4].

The neurophysiological phenomenon observed is the modulation of rhythmic activities (mu, beta and gamma rhythms) recorded over the sensorimotor cortex, termed as sensorimotor rhythm (SMR), encodes human motor tasks such as motor execution, motor intention, or motor imagery. In bilateral movement tasks, these modulations are termed as event-related (de-) synchronization (ERD/ERS) of mu (8-13 Hz) and beta (14-26 Hz) rhythms, indicating decrease in band powers recorded from contralateral sites and its increase in ipsilateral sites [1, 5]. Literature shows various representations of ERD/ERS using power spectral densities [6], time-frequency distributions by Short Time Fourier Transform [5] or Wavelet Transforms [7]. The goal of such studies are to identify most discriminative bands or time-frequency bins, from which the EEG features can be extracted to classify motor tasks.

Canonical Correlation Analysis (CCA) has been widely used in Steady State Visually Evoked Potential (SSVEP)-BCI to detect frequency components of EEG that characterizes stimulation frequencies and its harmonics [8, 9]. Many variants of CCA have been proposed to improve the accuracy and information transfer rate of BCI. CCA has been proposed in [10] to create spatial filters to improve Signal-to-Noise Ratio of EEG, and to analyze event related potentials (ERPs). In the current study, our objective is to identify the components of EEG that are highly correlated with imagined hand movement. We propose to use CCA to investigate spatio-spectral specific EEG activations underlying bilateral Motor Imagery (MI), and the feasibility of CCA-based features to classify imagined movement. CCA as a tool to determine discriminative time-frequency distributions of EEG activations in response to MI is also proposed. The study explores the scope of CCA in EEG analysis and for single-trial classification of MI in BCI.

The remaining paper is organized as follows: Section II describes the proposed methodology and its design in analysis and classification of EEG. Section III reports the results of our analysis. Section IV discusses the various results and its implications, followed by Section V that concludes our work.

II. PROPOSED METHODOLOGY

EEG based SMR-BCI research [4-7] focuses on various aspects of motor tasks such as: identification of its neural activation patterns, its classification using EEG-based features, and design of adaptive systems to decode these motor tasks. Due

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to the highly subject/task/session-specific nature of EEG activations, algorithms have been proposed to identify the discriminative information from spatial, spectral and temporal components of EEG. In this study, we propose the use of a CCA based analysis of EEG to identify the components of EEG that best correlates with a specific task. In addition, we use CCA to obtain distinct time-frequency representations of transformed EEG for each movement class. Compared to the existing methods, CCA based analysis considers the correlates of EEG with the user's response of the task, which allows it to automatically identify best spatial and spectral components of EEG.

A. Canonical Correlation Analysis (CCA) in MI

CCA is a multivariate statistical method that finds linear transformations of 2 datasets, which maximizes the correlation between the transformed datasets [11]. In the current study, we investigate EEG activations underlying bilateral MI, and the feasibility of using CCA in deriving features that can classify imagined movement. We define a subject-specific response for each mental state (motor imagery) and identify the significant spatial and spectral components of EEG that are highly correlated with this defined response.

Since, we are interested in both spatial and spectral components of EEG, we decompose the multi-channel EEG signal into various frequency bands. Let the EEG signal ($\in \mathbb{R}^{S \times T}$), recorded from C channels, decomposed to F frequency bands, where S is the number of samples per trial and T is the number of trials recorded for the subject. For each channel and frequency, we concatenate the trials, to obtain EEG signal, $X \in \mathbb{R}^{N \times P}$, such that the total number of samples is $N = S \times T$, and each sample in X is a vector of length $P = C \times F$.

As indicated earlier, CCA correlates EEG with known information like, stimulation frequencies in case of SSVEP. However, in case of MI, EEG needs to be correlated with patterns of elicited response. Literature has shown that the patterns of activation in EEG, are subject-specific, and hence require identifying the best channels and frequency bands that provide most information at a particular time instant, usually by

adaptive learning algorithms. Hence, creating a generic template, subject-specific response for bilateral MI activation is a major challenge.

We define user response template as trial averages of distinct spatial and spectral components, which is the class-specific trial averages of all the P components in X . However, from literature it is evident that channels in the sensory motor area, mostly C3, Cz and C4 contain more movement related information than the others. Hence, we use trial-averages from channel C3, from different frequency bands to create a template for Left and Right MI each $\in \mathbb{R}^{S \times Q}$ (Note that $Q = F$). For each trial in X , the corresponding template is assigned, and concatenated so as to create $Y \in \mathbb{R}^{N \times Q}$. In a similar way, we define another template $\hat{Y} \in \mathbb{R}^{N \times Q}$, but by assigning the mismatched class template for each trial. For example, for a trial in which subject perform left MI, Y contains left MI trial averaged response and $\hat{Y}$ contains right MI trial averaged response.

The objective of CCA, is to transform EEG signal ($X \in \mathbb{R}^{N \times P}$) and user response ($Y \in \mathbb{R}^{N \times Q}$), so as to maximize the correlation between the transformed signals. CCA determines, filters $W_X \in \mathbb{R}^{1 \times P}$ and $W_Y \in \mathbb{R}^{1 \times Q}$, such that, $U = W_X^T X$ and $V = W_Y^T Y$, has maximum correlation. Here, U and V are canonical variables corresponding to maximum canonical correlation. We also use a second CCA, which is to transform X and mismatched user response $\hat{Y}$, such that the correlation between transformed variables are minimum. Thus $\hat{W}_X \in \mathbb{R}^{1 \times P}$ and $\hat{W}_Y \in \mathbb{R}^{1 \times Q}$, are determined, such that, $\hat{U} = \hat{W}_X^T X$ and $\hat{V} = \hat{W}_Y^T \hat{Y}$, are least correlated. Here, $\hat{U}$ and $\hat{V}$ are canonical variables corresponding to lowest canonical correlation. The method is illustrated in Fig.1, where $X_p, Y_q, \hat{Y}_q$ are all N length vectors, and ρ and $\hat{\rho}$ are the corresponding correlation coefficients. We adopt this approach, so as to determine spatial and spectral components that maximizes correlation with the same class and also the ones that are least correlated with the opposite class.

The following sections explain how the above method is used in analysis and classification of the data.

B. Dataset and Experiment

To validate the proposed method, we used the publicly available dataset from an experiment conducted by the Wadsworth Center [12, 13]. In this, EEG recordings of 109 healthy volunteers were acquired from 64 channels using BCI2000. Data were sampled at a frequency of 160 Hz, and the EEG signals were referenced to earlobe electrodes. This data is available at <http://www.physionet.org/pn4/eeegmidb/>.

C. Experimental Analysis

The data corresponding to bilateral MI and rest prior to the MI are extracted from the first 85 subjects (Others are discarded due to inconsistent trial recording parameters). The EEG recorded during one second prior to movement cue and 4 seconds of MI are used in this analysis (indicated as -1 to 4 s). Thirty channels from motor and associated areas are identified and data from these are used for analysis. The data is pre-

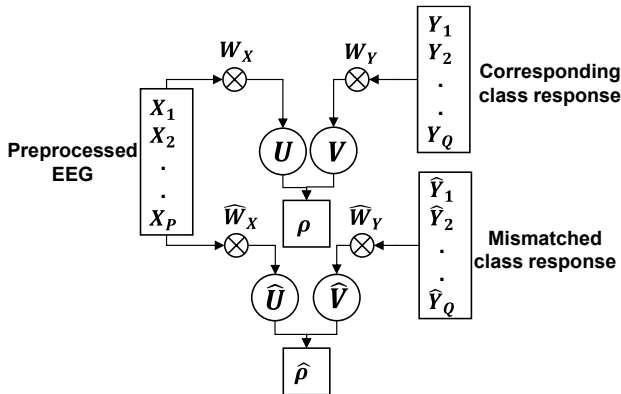


Figure 1. Proposed methodology using CCA.

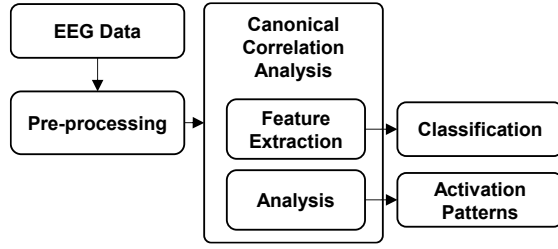


Figure 2. EEG signal processing.

processed by baseline correction. Baseline is defined as average of signal amplitudes from 1 second prior to movement cue, and this is subtracted from the entire trial. Further the signal is filtered from 2-40 Hz, using Chebyshev Type-II filter, to remove, powerline frequency and muscle artefacts. The pre-processed data is used in CCA analysis.

The flow of data in this offline processing method is given in Fig. 2. In order to validate the proposed approach of using CCA to find significant spectral and spatial components of EEG, we conducted various analyses on data. The main goal is to ensure that the transformed EEG signal using the above approach provide discriminative information regarding bilateral MI that can further be used in a BCI system. The CCA based method indicated in previous section is applied on 0.5 s windows of EEG, shifted by 0.4 s after each window. In each time window, weights are obtained for each spectral and spatial component, W_X and a transformed signal, U is obtained as a linear combination of these weights. We have defined frequency bands of 2 Hz width and 1 Hz overlap from 2-40 Hz ($F = 37$). The power spectral density of U in each time segment is calculated and time-frequency plots of transformed EEG is obtained. The time-frequency distribution, trial-averaged for each MI class is determined.

D. Single-trial Classification

For performance evaluation of the proposed method, a 5-fold cross validation is performed. The training data in each fold is used to define the inputs to CCA, and to determine the weight vectors $W_{X/Y}$ and $\hat{W}_{X/Y}$. For single-trial classification, each trial in the test set is transformed using W_X and $\hat{W}_X$, to obtain U and

$\hat{U}$ and correlated with the transformed templates, V and $\hat{V}$. For each trial, the correlation coefficients between $\{U, \hat{U}\}$ and $\{V, \hat{V}\}$ are obtained, which are saved as features per trial. Further, for the transformed EEG signals, U and $\hat{U}$, bandpowers are calculated and appended to the feature set. These six features per trial are further used for classification. We use a linear SVM classifier [14] and the performance reported is average classification accuracy over the cross-validation folds.

III. RESULTS

The results of the various analyses are reported in this section. We start with illustrating the functioning of CCA using spatio-spectral plots. Then we validate our hypotheses that CCA analysis of EEG can provide discriminative information regarding imagined bilateral hand movement, using time-frequency maps. Finally, the performance of EEG classification based on the proposed algorithm is reported in terms of classification accuracy (CA).

A. Analysis

CCA, if implemented as mentioned in Section II-A and II-C, enables to identify a filter, W_X , and a corresponding filtered signal, U . The following sections discuss their characteristics and significance.

1) Spatio-Spectral weights

Based on the definitions of our method, CCA will determine a filter W_X , whose each element will give a distinct weight to an EEG component from a specific single-channel and a filter band. This weight denotes the contribution of that specific spatio-spectral component to increase the correlation of EEG to the corresponding MI task. Fig. 3 reports, the weights determined in each 0.5s time window of the task, for Subject S7. It can be clearly seen that the weights are spatially localized (motor and nearby electrodes), and distributed along mu and beta bands. Also, the weights are temporally localized in (~0.6 to ~1.85s), which is the time segment in which the subject actively engages in the task. Subject, S7 is one of the best performing (CA>90 %) subjects and hence is chosen to demonstrate the CCA-based weights. For all the subjects, the weights are found to have similar trends in spectral and temporal information, but the channels that has high weights are found to be subject-specific.

2) Time-Frequency patterns

In order to demonstrate the discriminative behavior of the proposed feature, we use the time-frequency distribution (TFD) of filtered signal. In each time window, we obtain a filtered signal, U , for which power spectral density is determined. Similarly, TFD is calculated for all the trials for all subjects. The class-specific trial-averages for each subjects are then determined. This trial-averaged TFD is demonstrated using a contour plot for two specific subjects S70 and S83 in Fig. 4. The first row Fig.4 is for S70, where (a) denotes Left MI trial averages and (b) denotes Right MI trial averages. The discrimination can be clearly seen in 10-18 Hz, 1-2 second window. The second row in Fig.4 is for S83, in which (c) and

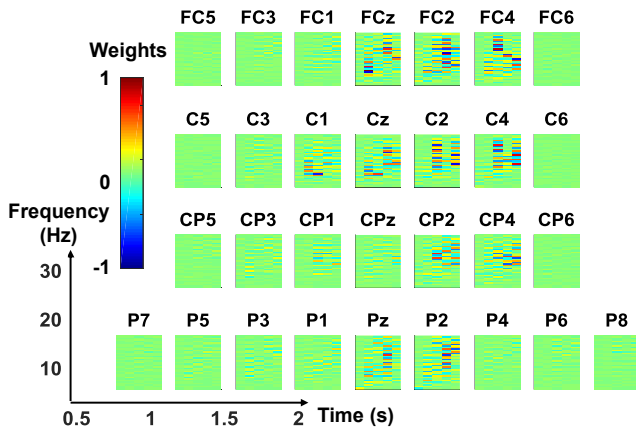


Figure 3. Temporal evolution of weights of spatio-spectral components obtained for subject, S7.

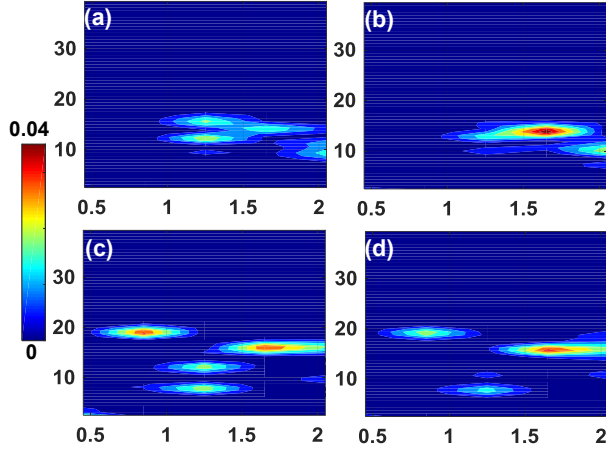


Figure 4. CCA based TFD for (a, b) Subject S70 and (c, d) subject S83. (a, c) are for trial averages of Left MI task. (b, d) are trial averages for Right MI task.

(d) are left and right MI trial averages respectively. It can be noted that, discrimination is less pronounced in this case.

These 2 subjects are specifically chosen based on classification performance (explained in the next section), in which S70, is a high performing and S83 is a low performing subject. These subjects are characterized by a distinct separable TFD (in S70) and overlapping TFD (in S83), which have in turn affected the respective classification performance.

Further, we have grouped the subjects into a set of 40 good performing subjects (Set-1: $CA \geq 60\%$) and another set of 45 subjects (Set-2: $CA < 60\%$). For each of these sets of average of TFDs are determined and shown in Fig. 5. Figs. 5(a, b) are for Set-1, (a) for Left MI and (b) for Right MI. Figs. 5(c, d) are for Set-2, (c) for Left MI and (d) for Right MI. It can be noticed that (1) even though variability between subjects have caused the distributed activation patterns, mu and beta band are still the most activated, (2) discrimination between classes can be observed in both Set-1 and Set-2, however, in mu band ($\sim 10\text{Hz}$), from 1 to 1.5 s, Set-1 appears to be more localized and distinct than Set-2.

B. Classification

The proposed features in Section II-D are used to classify right hand v/s left hand MI and the results of classification (CA) are reported in this section. Fig. 6 demonstrates the results as average and variance (error bars) over subjects' performance. Fig. 6 (a) shows the average over 85 subjects for each 2 second window of EEG data. (x-axis denotes starting point of window, x-axis: -1 denotes -1 to 1 s). An average accuracy of 60.56% is obtained, at the start of movement task and is maintained until 2.5 seconds of movement (x-axis: 0.5), beyond which performance falls. Hence, we used this time segment -1 to 2.5s to classify the tasks, and the results are reported in Fig. 6 (b). An average accuracy of 61.12% is obtained over 85 subjects. Out of 85 subjects, over 40 subject reported accuracy of more than 60% (Set-1 reported in Section IVA). An average accuracy of 73.34% with much less variance is observed in this case.

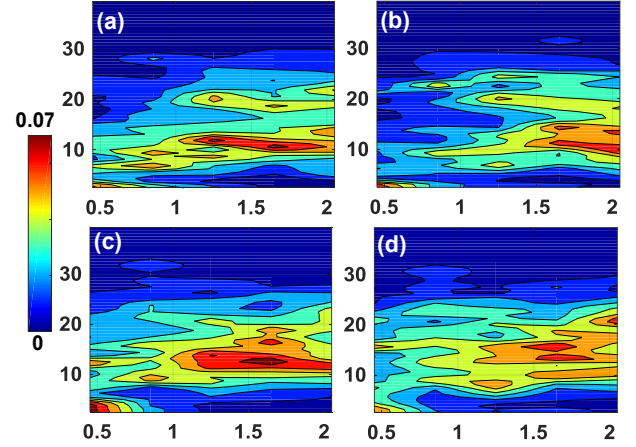


Figure 5. CCA based average TFD for (a, b) Set-1 and (c, d) Set-2. (a, c) are for trial averages of Left MI task. (b, d) are trial averages for Right MI task.

The variation in accuracy among subjects is quite high and hence, we created a distribution of subjects' performance v/s time window. The black markers indicate the number of subjects that reported high performance in the corresponding window, and blue bars indicate the average accuracy for the set of subjects in each window. As expected, the subjects whose CCA features are more discriminative during initial seconds of the task (specifically -1 to 2.5 s), provided accuracy of (68 to 72) %. This distribution also indicates that for few subjects, discriminative features become prominent in the later time segments, and for them classification accuracy is also comparatively lesser.

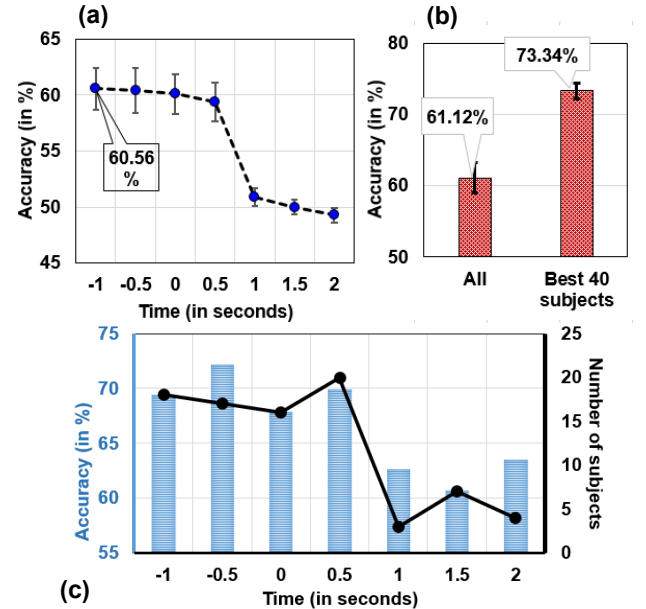


Figure 6. Performance of the proposed method in terms of classification accuracy.

IV. DISCUSSION

SMR-BCI researchers have widely investigated the ERD/ERS activation patterns and ways to quantify the same to define features that can be used to classify bilateral hand movement. The usual practice is to obtain time-frequency activation maps in C3, Cz or C4 and identify the characteristic patterns in mu and beta band, distinctively for right and left hand motor execution or imagination [5]. In this study, we used a CCA-based approach to obtain spatio-spectral weights for EEG, such that the transformed EEG is highly correlated with responses specific to MI. The time-frequency distribution of the transformed EEG is then obtained in a similar attempt to see discriminative task-specific patterns.

Figures 4 and 5 show the EEG representations derived using the proposed approach. The distinct patterns can be clearly observed in the mu and beta bands, at the onset of movement. Based on the template used, the proposed method, identifies the most informative spatial and spectral components, and weights them accordingly into a linear transformation of EEG, for which we can determine TFD. Thus this method provides an alternate method of analysis of TFD which consolidates information in the various electrodes.

Further, the accuracy reported in Fig. 6, validates the use of the proposed approach for single-trial classification of right v/s left hand motor imagery. Previous reports of MI classification on the same data reported 60-63% in [15], for various subset of channels using channel selection algorithms and filter bank common spatial pattern based features. Various other studies [16, 17] have reported average accuracies for a specific set of best scoring subjects based on their methods. The accuracy achieved using the proposed approach are comparable with the existing reports. This confirms the feasibility of using CCA based features for EEG analysis.

In summary, we have proposed to use CCA to identify the most informative spatio-spectral components of EEG, create a transformed EEG signal, and use the same to create discriminative TFD to classify bilateral MI. The proposed methodology has scope of investigation in two major factors: (1) We have defined the user responses and inputs to CCA, using trial-averages of single-channel EEG (C3). This, of course, is variable, and we can use any other channel or even subset of channels or a simulated signal to define user responses to MI. We are yet to identify how these can influence the performance of classification. (2) We have adopted a parallel implementation of two CCA stages, one which aims to maximize correlation with the corresponding mental state, and another to minimize correlation with the mismatched mental state. An immediate modification could be to explore ways to define a transformation that can uncorrelate the transformed EEG and response, rather than to minimize correlation. These two factors can greatly contribute to optimizing the performance of the method as well as to improve the classification accuracy of the features used.

V. CONCLUSION

In this paper, we propose a novel approach based on CCA to classify bilateral hand movement imagination. CCA is also used to obtain time frequency distribution that shows significant differences between right and left hand MI. An average accuracy

of 61.12% and 73.34% is obtained over 85 subjects and over 40 subjects with CA ($\geq 60\%$) respectively. The results indicate the scope of CCA based transformations of EEG in analysis and classification of imagined hand movement.

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